Image 1

Recurrence Relation and Generating Functions

There are some basic techniques of counting, such as permutation, combination, etc. Many counting problems can not be solved by using the basic techniques of counting. Such problems can be solved by recurrence relation and generating function.

📌 Definition of Recurrence Relation

A **recurrence relation** for the sequence a_n is an equation that expresses a_n in terms of one or more of the previous terms of the sequence, namely $a_0, a_1, a_2, \ldots, a_n$ for all integer n with $n \geq n_0$, where n_0 is a non-negative integer.

Example:

$$a_n = a_{n-1} - a_{n-2}$$

> The order of this equation is $n-(n-2)=2$

📌 Solving Recurrence Relations

- Basically, when solving such recurrence relations, we try to find solutions of the form $a_n = r^n$, where r is a condition constant.

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- $a_n = r^n$ is a solution of the recurrence relation.

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

if and only if

$$r^n = c_1 r^{n-1} + c_2 r^{n-2} + \cdots + c_k r^{n-k}$$

- Divide this equation by r^{n-k} and subtract the right-hand side from the left:

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \cdots - c_{k-1} r - c_k = 0$$

This is called the **characteristic equation** of the recurrence relation.

- The solutions of this equation are called the **characteristic roots** of the recurrence relation.
- Let us consider **linear homogeneous recurrence relations of degree two**.

Theorem

Let c_1 and c_2 be real numbers. Suppose that $r^2 - c_1 r - c_2 = 0$ has two distinct roots r_1 and r_2.

Then the sequence a_n is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n \quad \text{for } n = 0, 1, 2, \ldots$$

where α_1 and α_2 are constants.

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📌 Solution of a Recurrence Relation

A sequence is called a **solution** of a recurrence relation if it satisfies the recurrence relation.

Example: The sequence a_n with $a_n = 3n$ is a solution of the recurrence relation:

$$a_n = 2a_{n-1} - a_{n-2}, \quad \forall\, n \geq 2 \tag{1}$$

Solution: The given recurrence relation is:

$$a_n = 2a_{n-1} - a_{n-2}$$

We know $a_n = 3n$, so:

$$a_0 = 3 \cdot 0 = 0, \quad a_{n-1} = 3(n-1), \quad a_{n-2} = 3(n-2)$$

The right-hand side of equation (1):

$$\begin{gathered}\text{R.H.S.} = 2a_{n-1} - a_{n-2} \\ = 2 \cdot 3(n-1) - 3(n-2) \\ = 6n - 6 - 3n + 6 \\ = 3n \\ = a_n \\ = \text{L.H.S.}\end{gathered}$$

$\therefore$ $a_n = 3n$ is the solution of the recurrence relation $a_n = 2a_{n-1} - a_{n-2}$.

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📌 Problem

Someone deposits $10{,}000$ Tk in a savings account at a Bank yielding 5 per year with interest compounded annually. How much money will be in the account after 30 years?

Solution: Let P_n be the amount in account after n years. We can derive the following recurrence relation:

$$P_n = P_{n-1} + 0.05, P_{n-1}$$

$$\Rightarrow P_n = P_{n-1}(1 + 0.05)$$

$$\Rightarrow P_n = (1.05), P_{n-1} \tag{1}$$

Now, initial deposit $P_0 = 10{,}000$ Tk.

$$P_1 = (1.05), P_0 = (1.05) \times 10{,}000$$

$$P_2 = (1.05), P_1 = (1.05)(1.05) \times 10{,}000 = (1.05)^2 \times 10{,}000$$

$$\therefore P_{30} = (1.05)^{30} \times 10{,}000 \text{ Tk}$$

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📌 Linear Homogeneous Recurrence Relations

Problem: What is the solution of the recurrence relation

$$a_n = a_{n-1} + 2a_{n-2}$$

with $a_0 = 2$ and $a_1 = 7$?

Solution: Given that,

$$a_n = a_{n-1} + 2a_{n-2} \tag{1}$$

Let $a_n = r^n$ be a solution of equation (1).

$$\therefore r^n = r^{n-1} + 2r^{n-2}$$

Dividing both sides by r^{n-2}:

$$\frac{r^n}{r^{n-2}} = \frac{r^{n-1}}{r^{n-2}} + 2\frac{r^{n-2}}{r^{n-2}}$$

$$\Rightarrow r^2 = r + 2$$

$$\Rightarrow r^2 - r - 2 = 0$$

$$\Rightarrow r^2 - 2r + r - 2 = 0$$

$$\Rightarrow r(r-2) + (r-2) = 0$$

$$\Rightarrow (r+1)(r-2) = 0$$

$$\therefore r = 2, \quad -1$$

$\therefore$ Roots are real and distinct.

$\therefore$ The general solution is:

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n = \alpha_1 \cdot 2^n + \alpha_2 \cdot (-1)^n \tag{2}$$

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When $n = 0$:

$$a_0 = \alpha_1 \cdot 2^0 + \alpha_2 \cdot (-1)^0$$

$$\Rightarrow 2 = \alpha_1 + \alpha_2 \qquad [\because a_0 = 2]$$

$$\therefore \alpha_1 + \alpha_2 - 2 = 0 \tag{3}$$

When $n = 1$:

$$a_1 = \alpha_1 \cdot 2^1 + \alpha_2 \cdot (-1)^1$$

$$\Rightarrow 7 = 2\alpha_1 - \alpha_2 \tag{4}$$

$$\Rightarrow 2\alpha_1 - \alpha_2 - 7 = 0$$

Now solving (3) and (4):

$$\frac{\alpha_1}{-7-2} = \frac{\alpha_2}{-4+7} = \frac{1}{-1-2}$$

$$\Rightarrow \alpha_1 = \frac{-9}{-3}, \quad \alpha_2 = \frac{3}{-3}$$

$$\therefore \alpha_1 = 3, \quad \alpha_2 = -1$$

Putting the values of α_1 and α_2 in equation (2):

$$\boxed{a_n = 3 \cdot 2^n - (-1)^n}$$

which is the solution of the given recurrence relation.

Problem: What is the solution of the recurrence relation

$$a_n = -6a_{n-1} - 9a_{n-2}, \quad n \geq r$$

with $a_0 = -6$ and $a_1 = 3$?

Solution: Given that,

$$a_n = -6a_{n-1} - 9a_{n-2} \tag{1}$$

Let $a_n = r^n$ be a solution of equation (1).

$$\therefore r^n = -6r^{n-1} - 9r^{n-2}$$

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Dividing both sides by r^{n-2}, we get:

$$r^2 = -6r - 9$$

$$\Rightarrow r^2 + 6r + 9 = 0$$

$$\Rightarrow (r+3)^2 = 0$$

$$\Rightarrow (r+3)(r+3) = 0$$

$$\therefore r = -3,\ -3$$

$\therefore$ The roots are real and **equal**.

$\therefore$ The general solution is:

$$a_n = \alpha_1 r_1^n + n\alpha_2 r_2^n \tag{2}$$

When $n = 0$:

$$a_0 = \alpha_1 + 0$$

$$\Rightarrow -6 = \alpha_1 + 0$$

$$\Rightarrow \alpha_1 + 6 = 0 \tag{3}$$

$$\therefore \alpha_1 = -6$$

When $n = 1$:

$$a_1 = \alpha_1 r_1 + \alpha_2 r_2$$

$$\Rightarrow 3 = (-6)(-3) + \alpha_2(-3)$$

$$\Rightarrow 3 = 18 - 3\alpha_2$$

$$\Rightarrow 3\alpha_2 = 15$$

$$\therefore \alpha_2 = 5$$

Putting the values of α_1 and α_2 in equation (2):

$$a_n = -6r_1^n + 5n, r_2^n$$

$$= -6(-3)^n + 5n(-3)^n$$

which is the solution of the given recurrence relation.

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Problem: What is the solution of the recurrence relation

$$a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$$

where $a_0 = 2,\ a_1 = 5,\ a_2 = 15?$

Solution: Given that,

$$a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3} \tag{1}$$

Let $a_n = r^n$ be a solution of equation (1).

$$\therefore r^n = 6r^{n-1} - 11r^{n-2} + 6r^{n-3}$$

Dividing both sides by r^{n-3}, we get:

$$r^3 = 6r^2 - 11r + 6$$

$$\Rightarrow r^3 - 6r^2 + 11r - 6 = 0$$

$$\Rightarrow r^3 - r^2 - 5r^2 + 5r + 6r - 6 = 0$$

$$\Rightarrow r^2(r-1) - 5r(r-1) + 6(r-1) = 0$$

$$\Rightarrow (r-1)(r^2-5r+6)=0$$

$$\Rightarrow (r-1)(r^2-3r-2r+6)=0$$

$$\Rightarrow (r-1)r(r-3)-2(r-3)=0$$

$$\Rightarrow (r-1)(r-2)(r-3)=0$$

$$\therefore r=1,\ 2,\ 3$$

Here, the roots are real and distinct.

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The general solution is:

$$a_n=\alpha_1 r_1^n+\alpha_2 r_2^n+\alpha_3 r_3^n \tag{2}$$

When $n=0$:

$$a_0=\alpha_1+\alpha_2+\alpha_3 \qquad [\because a_0=2]$$

$$\Rightarrow 2=\alpha_1+\alpha_2+\alpha_3$$

$$\Rightarrow \alpha_1+\alpha_2+\alpha_3-2=0 \tag{3}$$

When $n=1$:

$$a_1=\alpha_1 r_1+\alpha_2 r_2+\alpha_3 r_3 \qquad [\because a_1=5]$$

$$\Rightarrow 5=\alpha_1 r_1+\alpha_2 r_2+\alpha_3 r_3$$

$$\Rightarrow \alpha_1 r_1+\alpha_2 r_2+\alpha_3 r_3-5=0 \tag{4}$$

$$\Rightarrow \alpha_1+2\alpha_2+3\alpha_3-5=0 \tag{4}$$

When $n=2$:

$$a_2 = \alpha_1 r_1^2 + \alpha_2 r_2^2 + \alpha_3 r_3^2 \qquad [\because a_2 = 15]$$

$$\Rightarrow 15 = \alpha_1 r_1^2 + \alpha_2 r_2^2 + \alpha_3 r_3^2$$

$$\Rightarrow \alpha_1 r_1^2 + \alpha_2 r_2^2 + \alpha_3 r_3^2 - 15 = 0 \tag{5}$$

$$\Rightarrow \alpha_1 + 4\alpha_2 + 9\alpha_3 - 15 = 0 \tag{5}$$

$(4) - (3)$: $\quad \alpha_2 + 2\alpha_3 - 3 = 0$

$(5) - (3)$: $\quad 3\alpha_2 + 8\alpha_3 - 13 = 0$

Solving (6) and (7), we get:

$$\frac{\alpha_2}{-26 + 24} = \frac{\alpha_3}{-9 + 13} = \frac{1}{8 - 6}$$

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$$\Rightarrow \frac{\alpha_2}{-2} = \frac{\alpha_3}{4} = \frac{1}{2}$$

$$\therefore \alpha_2 = -1, \quad \alpha_3 = 2$$

From (3): $\quad \alpha_1 - 1 + 2 - 2 = 0$

$$\therefore \alpha_1 = 1$$

$$\therefore \alpha_1 = 1, \quad \alpha_2 = -1, \quad \alpha_3 = 2$$

$\therefore$ The general solution is:

$$a_n = 1 \cdot (1)^n + (-1)(2)^n + (2)(3)^n$$

$$\boxed{\therefore a_n = 1 - 2^n + 2 \cdot 3^n}$$

which is the required solution of the given recurrence relation.

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Problem: Find the solution of the recurrence relation

$$f_n = f_{n-1} + f_{n-2}, \quad f_0 = 0,\ f_1 = 1$$

Find the explicit formula for the Fibonacci sequence/number.

Solution: Given that,

$$f_n = f_{n-1} + f_{n-2} \tag{1}$$

Let $f_n = r^n$ be a solution of equation (1).

Now, $r^n = r^{n-1} + r^{n-2}$

Dividing both sides by r^{n-2}:

$$r^2 = r + 1$$

$$\Rightarrow r^2 - r - 1 = 0$$

$$\Rightarrow r = \frac{1 \pm \sqrt{1+4}}{2} \qquad \left[\because x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}\right]$$

$$\therefore r = \frac{1+\sqrt{5}}{2}, \quad \frac{1-\sqrt{5}}{2}$$

$\therefore$ The roots are real and distinct.

$\therefore$ The general solution is:

$$f_n = \alpha_1 r_1^n + \alpha_2 r_2^n \tag{2}$$

When $n = 0$:

$$f_0 = \alpha_1 + \alpha_2 \qquad [\because f_0 = 0]$$

$$\Rightarrow 0 = \alpha_1 + \alpha_2$$

$$\therefore \alpha_1 + \alpha_2 = 0 \tag{3}$$

$$\therefore \alpha_1 = -\alpha_2$$

When $n = 1$:

$$f_1 = \alpha_1 r_1 + \alpha_2 r_2 \qquad [\because f_1 = 1]$$

$$\Rightarrow 1 = \alpha_1 r_1 + \alpha_2 r_2$$

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$$\Rightarrow \alpha_1 r_1 + \alpha_2 r_2 - 1 = 0 \tag{4}$$

Solving (3) and (4):

From (4): $\quad -\alpha_2 r_1 + \alpha_2 r_2 - 1 = 0$

$$\Rightarrow \alpha_2 (r_2 - r_1) - 1 = 0$$

$$\Rightarrow \alpha_2 = \frac{1}{r_2 - r_1}$$

$$= \frac{1}{\dfrac{1-\sqrt{5}}{2} - \dfrac{1+\sqrt{5}}{2}}$$

$$= \frac{1}{\dfrac{1-\sqrt{5}-1-\sqrt{5}}{2}} = \frac{2}{-2\sqrt{5}} = -\frac{1}{\sqrt{5}}$$

$$\therefore \alpha_2 = -\frac{1}{\sqrt{5}} \quad \text{and} \quad \alpha_1 = \frac{1}{\sqrt{5}}$$

From (2):

$$f_n = \frac{1}{\sqrt{5}}, r_1^n - \frac{1}{\sqrt{5}}, r_2^n$$

$$\therefore f_n = \frac{1}{\sqrt{5}}(r_1^n - r_2^n) \quad \text{(solution of the given recurrence relation)}$$

$$= \frac{1}{\sqrt{5}}\left\{\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right\}$$

$$\boxed{\therefore f_n = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n}$$

which is the required formula. ■